

**Q1. Need for Server-Side Validation** despite client-side validation: Client-side validation can be bypassed (disabled JavaScript, modified requests via Postman/curl), so server-side validation ensures **data integrity, security, and consistency** at the database level. It protects against malicious attacks (SQL injection, XSS), enforces business rules that client can't know (e.g., unique username checks), and provides a final safety layer before data persistence.

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**Q2. JSR-380 (Bean Validation 2.0) Annotations:**

- **@NotNull** – Ensures field is not null (accepts empty string, ""). *Example: @NotNull private String name;*
- **@Size(min=2, max=30)** – Validates string/collection/array length is between min and max. *Example: @Size(min=8, max=20) private String password;*
- **@Min(value=18)** – Validates numeric value is at least the specified minimum (inclusive). *Example: @Min(18) private int age;*
- **@Max(value=100)** – Validates numeric value is at most the specified maximum (inclusive). *Example: @Max(100) private double percentage;*
- **@Pattern(regexp="^[A-Za-z]\*\$")** – Validates field matches a specific regular expression. *Example: @Pattern(regexp="^[0-9]{10}\$") private String phoneNumber;*
- **@Email** – Validates field is a well-formed email address (supports RFC 5322). *Example: @Email private String email;*